SCORE: 2022
HIGH-SCORE: 2022



LET'S PLAY CHESSBOT!

MENU

SIGN IN

Harwdare

ChessBot combines a range of electrical, electronic and mechanical components to ensure reliable operation and precise movement.
The system is powered by three power supplies providing 48 V, 24 V and 5 V, supplied by MEAN WELL and PHOENIX CONTACT. Three SHS motor drives control the three AEC stepper motors, which are responsible for moving the Cartesian robot along its axes.
The electrical cabinet also houses a Siemens miniature circuit breaker and switch, providing the main electrical protection and switching functions.
The system is managed by a Siemens PLC, which coordinates the robot's movements and the communication between the various components. A Siemens KTP700 Basic HMI panel provides the user interface, allowing the operator to interact with the system and control its operation.
Finally, a DH-ROBOTICS electrically actuated gripper, powered at 24 V DC, handles the chess pieces during the game.
Technical Notes

- Check the power supplies' output voltages after extended periods of inactivity and adjust them if necessary.
- Ensure adequate ventilation inside the electrical cabinet, as the motor drives can reach high temperatures.
- Install the electrical cabinet in an environment with acceptable humidity levels.
- Never introduce foreign objects into the electrical cabinet while the system is running.

Software

The following software tools were used in the design, development and testing of the Cartesian robot:

- PRONETA
- TIA Portal V17
- Visual Studio Code

PRONETA, a tool for PROFINET network commissioning and diagnostics, was used to configure the network, enabling communication between the various devices.
The PLC was programmed using TIA Portal V17 and the KOP (Ladder Logic) and SCL programming languages. It manages axis movements, communication with the HT7 drives via the PROFINET protocol, limit switch handling and control of the electrically actuated gripper.
The program also includes comments to ensure that the selected algorithms and processes can be clearly understood.
Visual Studio Code was used to run the Sunfish chess engine, implemented in Python, to validate the player's moves and calculate the engine's response to each opponent's move.

Interface



The user interface features an engaging 1980s video game-inspired design, with a clear layout that allows users to quickly understand and operate the system.
The left side of the HMI panel is dedicated to interacting with the Cartesian robot, while the right side displays the chessboard used to play the game.

- 1: System enable button.
- 2: System start button, which initiates the homing routine and enables the player to make the first move.
- 3: Stop button, which immediately stops the motors and resets the game to its initial state.
- 4: Indicator lights.
- 5: Notification showing whose turn it is.
- 6: Chessboard. To make a move, the player must select the starting square and then the destination square. Invalid moves are indicated by a flashing signal, prompting the player to select a new move.